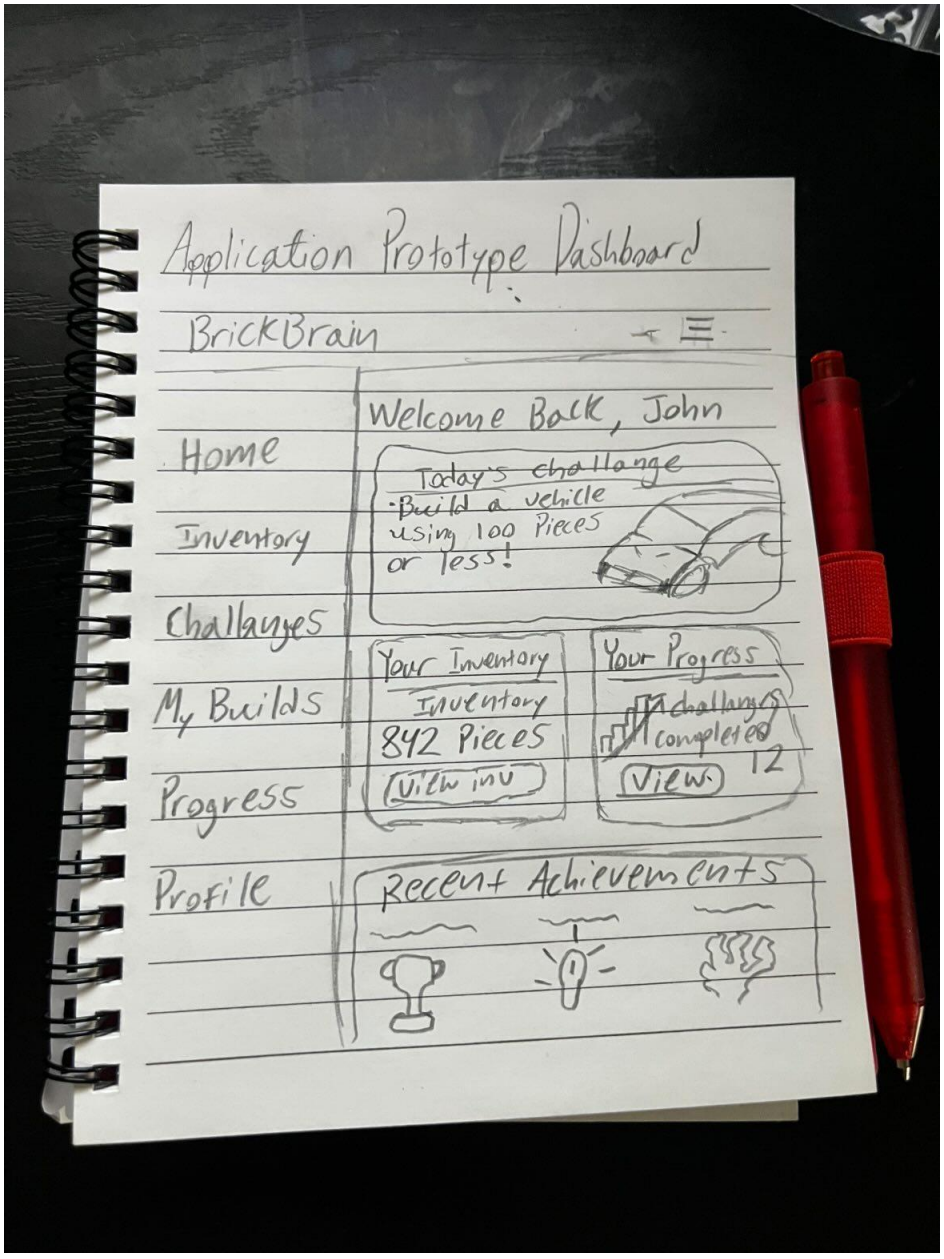
BrickBrain: Intelligent LEGO Build System

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Technical Field

This project mainly works around Software Engineering, web application development, user interface design, and inventory data management. The current prototype uses HTML, CSS, and JavaScript. Adaptive challenge logic is put into use through user-selected settings and feedback, that has more advanced AI capabilities remain in a longer-term direction.

Background Information

LEGOs and building them is a very big hobby for many people all over the world. A hobby that promotes creativity, spatial rationing, and engineering-based problem solving as well. Many LEGO enthusiasts tend to gather lots of LEGO sets and proceed to build a large collection. Some of these collections are often underutilized because a majority of the time people find it difficult to identify new build possibilities with a lot of left over and extra parts.

Many current LEGO tools focus too much on providing official set instructions or trying to provide other designs with digital environments or fields. Even though these are useful, they aren't as effective in helping users discover new ideas and challenges to open the creative door towards builds based off their inventory.

This project was inspired by my passion for LEGOs and wanting to help other enthusiasts like myself to help others interact with their existing collections and create challenges for themselves to improve their skill in problem solving and creativity. The goal is to use artificial intelligence to transform this into a dynamic and more creative opportunity for users while also encouraging an innovative and experimentation in design.

Prior Art (Research)

Prior Art

Brickit

- Brickit is a mobile application that helps LEGO users identify pieces from their collection and provides suggestions for models they can build using those available parts. The platform focuses on inventory analysis and matching existing designs to a user's collection.

How BrickBrain Differs

- BrickBrain differs from Brickit because it doesn't focus on recommending existing models. Instead, BrickBrain functions as an adaptive LEGO design coach system that generates personalized build challenges based on a user's inventory, interests, and skill level. The goal is to push original creations and skill development rather than just simply matching pieces a user has to pre-existing designs.

Citation

Brickit. (2025). Brickit App. https://brickit.app

Rebrickable

- Rebrickable is an online LEGO inventory and database application that allows users to track their collections, discover existing builds, and help identify missing pieces required for specific builds. ReBrickable focuses on inventory managing and finding MOCs or official sets.

How BrickBrain Differs

- BrickBrain differs from Rebrickable because its primary purpose isn't inventory tracking or finding existing builds. Instead, BrickBrain creates adaptive building challenges that help encourage users to create original LEGO designs using their available pieces on a build. By combining the options of themes, difficulty levels, and challenge requirements, BrickBrain promotes creativity, experimentation, and skill growth rather than simply providing model recommendations.

Citation

Rebrickable. (2025). Rebrickable LEGO Database. https://rebrickable.com

Project Description

BrickBrain is a LEGO design coach system that generates build challenges that are based on a user's inventory, interests, and skill level. Users then have the option to choose the difficulty based off their experience, rather than simply recommending existing LEGO models, the system encourages users to create original builds by providing creative constraints, themes, and objectives soley based off the pieces they have in their system.

BrickBrain combines inventory management, AI, and challenge generation to transform a LEGO collection into interactive learning and creativity experiences that would help a user’s skill development. By analyzing a user's available pieces and preferences, BrickBrain will create building activities and challenges that’ll create problem-solving, experimentation, and design thinking.

This project targets LEGO hobbyists, collectors, students, and creative builders who want to improve their building skills while making greater use of their existing collections. BrickBrain achieves this by providing personalized challenges and guided creativity, BrickBrain helps users discover new ways to build and learn rather than simply following instructions from existing sets. This opens many opportunities for collectors and people who want to learn and challenge themselves.

Innovation Claim

The innovation BrickBrain gives is that it functions as an adaptive LEGO design coach rather than just being a traditional inventory system or build recommendation application. Many existing applications primarily focus on tracking collections, displaying premade instructions, or recommending existing models. BrickBrain cracks this as it generates personalized build challenges based on a user's inventory, interests, and skill level to encourage original creations to help build their skill development while using BrickBrain.

BrickBrain promotes creativity by giving users the options to choose their difficulty as well as themes and design requirements. By putting both inventory analysis with adaptive challenge generation together, this helps BrickBrain create a learning-focused experience that helps users develop building skills while encouraging experimentation and innovation all together in one place.

Usage Scenario

A user opens up BrickBrain and proceeds to start importing their past sets and some pieces they have laying around. The system then stores and organizes the inventory, allowing the user to select their preferred themes, interests, and challenge difficulty levels to give the system a preference of the user.

BrickBrain analyzes the pieces and breaks down the sets that were imported which then the system would generate a personalized build challenge for the user. For example, a user may receive a challenge to create a city themed vehicle using fewer than 125 pieces, including only 3

wheels. The challenge is specifically designed around the user's inventory and selected preferences.

As the users complete new challenges, and add feedback, they can keep track of their progress and keep receiving new challenges push to encourage users on new and unique building techniques as well as helping with creating thinking. BrickBrain serves as both an intelligent inventory management tool and a personalized LEGO creativity coach that promotes original design and skill development.

SIP BRIEF PART 2

Evaluation Criteria

1. Can users successfully access and navigate the BrickBrain platform?
2. Can users manually add LEGO pieces to their inventory?
3. Can users browse and manage their LEGO inventory?
4. Can users edit or remove inventory items when necessary?
5. Can users select a preferred building theme?
6. Can users select a challenge difficulty level?
7. Can the system generate personalized building challenges?
8. Do generated challenges include clear objectives and requirements?
9. Can users view current and previous challenges?
10. Can users track completed challenges and progress?
11. Does the challenge generation system encourage original LEGO creations?
12. Is the user interface easy to understand and navigate?
13. Does the system function without major errors during testing?
14. Are challenge requirements generated based on user preferences and inventory information?
15. Does the completed prototype demonstrate the concept of an adaptive LEGO design coach?

Goals and Tasks Associated with the Project

Objective 1: Developing the Inventory Management System

The Inventory Management System has been developed to let users organize and manage their LEGO collections. The remaining work is to connect stored inventory information to the Adaptive Challenge System and combine the two core features.

Tasks:

- Research inventory storage methods for LEGO collections.
- Design the database structure used to store LEGO pieces and inventory information.
- Create inventory browsing functionality.
- Develop inventory addition, editing, and deletion features.
- Test inventory storage and retrieval functionality.
- Inventory Management System developed, integration with Adaptive Challenges and final testing are ongoing (November 2026).

Objective 2: Develop the Adaptive Challenge Generation System

The Adaptive Challenge System is functional. Users can generate challenges, choose or customize difficulty and challenge types, and submit feedback that adjusts to the user and changes how their future challenges work based on the feedback.

Tasks:

- Research challenge-based learning and creativity systems.
- Design challenge generation logic.
- Create theme selection functionality.
- Create difficulty selection functionality.
- Develop challenge requirements, categories, and rules.
- Test generated challenges for usability and creativity.
- Core Adaptive Challenge System developed, inventory integration and final testing are ongoing (November 2026).

Objective 3: Develop the User Interface

This is a web-based interface that includes a dashboard and has a separate inventory, and challenge systems that are core features of the application. Current work focuses on connecting these systems together, improving navigation and usability, and preparing the prototype for final testing.

Tasks:

- Design wireframes and layouts for pages on the application.
- Create inventory management screens.
- Design challenge display screens.
- Develop navigation and user interaction features.
- Have testing done and gather feedback.
- Implement interface improvements based on testing results.

Objective 4: Test and Evaluate the Prototype

The last objective is to make sure that all the major components function correctly and satisfy project goals.

Tasks:

- Test inventory management functionality.
- Test challenge generation functionality.
- Conduct user testing sessions.
- Document bugs and software issues.
- Implement corrections and improvements.
- Evaluate project success using the evaluation criteria established above.

Description of Design Prototype

BrickBrain is being developed as a web-based prototype using HTML, CSS, and JavaScript. Its current features include a dashboard, an Inventory Management System, and a functional Adaptive Challenge System. The inventory and challenge components are still being connected.

The Inventory Management System has been developed for organizing LEGO pieces, while the Adaptive Challenge System allows users to select difficulty and challenge settings. The remaining work is to verify the integration between inventory information and generated challenges.

The Adaptive Challenge System right now helps to create challenges from user-selected settings and adapts difficulty recommendations based on feedback. Using the stored inventory to shape challenge requirements is the next major development goal.

The prototype currently has a functional dashboard, inventory management features, and adaptive challenge generation. The next steps are to connect and make sure that the inventory data with the challenge logic work all together in order to test the full workflow and change the interface before the final showcase.

SIP Brief Part 3 – to be filled out in SIP409

Evaluation Plan

The main evaluation plan for BrickBrain focuses on testing the main core features of the application and making sure to test everything and make sure it all works so it can be the best and polished as it can be. The goal is to determine whether BrickBrain successfully works as an adaptive LEGO design coach that helps generate personalized build challenges based on user inventory, interests, and skill level.

The first part of testing will mainly focus on the core features like the adaptive challenge systems and the inventory system as these two are the two main features of BrickBrain. I will test whether users can add, edit, remove, and organize their LEGO pieces. For the challenge system, I will make sure to test the different difficulty levels as I find it important for those levels to be very accurate. I will also test the feedback system to make sure that it works with the two core systems to make it more immersive for the users.

The second part of testing will be how the two main systems work together. Since BrickBrain is designed to generate challenges based on the users' available LEGO pieces, I will test the different inventories to see whether the challenges match the available pieces and the right selected difficulty. I will also make sure to test the user interface heavily. As for me, it's important that users have easy access to the features and make things easy for them.

Finally, I plan to gather as much feedback as I can from others to identify any bugs or anything that they might dislike and improve anything that is out of order. I will compare testing results and any additions that might be needed as well. My big goal is to have a clean and polished working prototype that helps show BrickBrain's main features and purpose of encouraging creativity for users that are personalized through build challenges.

Project Completion Assessment

Through my time working on BrickBrain, I've made a large amount of progress and learned many new ways of processing things and breaking things down. My biggest accomplishment I'm super proud of is my completion on the adaptive challenge system, which worried me early on, especially trying to get the inventory system to work with it makes it a little more nerving and exciting. I'm just proud of how these features have turned out and how much I've done and learned through my SIP progress.

One of the things that went really well was the development of the adaptive challenge system, which allows the users to choose different difficulties and provide user feedback on their challenges. The system uses this feedback to adjust to any future recommendations. I've also

developed the Inventory system which still has some flaws that I'm working on, but it will allow users to manage their LEGO pieces so that the other main core system can use it. I'm still working on making the two systems work together properly.

One of the biggest challenges I've faced so far is figuring out how to connect the main features of BrickBrain together. I've learned that developing the systems individually is one thing but making them work together can be a much bigger thing. This project has taught me how important planning is and this is one that I've tried to avoid a bit and it's cost me, but it's still going good so far.

If I were to start this project again, I would spend more time planning on how the different systems would connect before developing them both. I would also begin the testing much earlier than later to avoid running into any issues and saving me time. Overall, BrickBrain has been challenging for the better. It's been an enjoyable experience, and I'm looking forward to finishing the remaining features and preparing for the final showcase.

Appendices

Use the following format:

Appendix A: BrickBrain initial concept sketch – BrickBrain Sketch.jpg

Appendix B: BrickBrain prototype dashboard – Dashboard.png

Appendix C: Adaptive Challenge System – Challenge.png

Appendix D: BrickBrain user interface design – BrickBrain.png

Appendix E: BrickBrain logo – BrickBrain logo.png